

Omninet_AI_System_Requirements_Document_v1.0

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010 Functionality System Requirements

The capability of the system to provide functions which meet stated and implied needs when the software is used under specified conditions.

ID	Ver	MRD Ref	Description	Priority*
010-010	Suitability			
010-010-010	1.0	3.1	Kiosk design shall be suitable for installation in public places (airports, malls, hotels, etc.), with robust, weather-resistant enclosures where required.	1
010-010-020	1.0	3.1	Kiosks shall allow access to all publicly available Internet sites through standard URLs, subject to content control policies.	1
010-010-030	1.0	3.1	Kiosks shall provide a modern touchscreen interface, supplemented by an optional on-	1

ID	Ver	MRD Ref	Description	Priority*
			screen keyboard and voice input where supported by the AI Assistant.	
010-010-040	1.0	3.1	The AI Support Center dashboard shall provide agents with a web-based graphical user interface, accessible via modern browsers, with real-time updates.	2
010-010-110	1.0	3.1.2	Kiosks shall accept payment in local currencies, including cash (bills and coins), credit/debit cards (with chip and NFC), and digital wallets (Apple Pay, Google Pay, QR codes).	2
010-010-210	1.0	3.1.5	Kiosks shall be configured to interact with customers in the primary local language	1

ID	Ver	MRD Ref	Description	Priority*
			by default, with the option to switch via the AI Assistant.	
010-010-220	1.0	3.1.5	At the Welcome screen, kiosks shall allow users to select a secondary local language. Once selected, the core UI and AI Assistant shall interact in that language until session termination.	2
010-010-230	1.0	3.1.5	All kiosk web browsers shall support Unicode and display web pages in all languages supported by the underlying OS and browser engine, with the AI Assistant providing real-time translation if needed.	2

ID	Ver	MRD Ref	Description	Priority*
010-010-240	1.0	3.1.5	The AI Assistant shall support real-time conversational translation for at least 20 major languages, including English, Spanish, French, Mandarin, Arabic, Hindi, etc.	3
010-010-250	1.0	3.1.5	The AI Assistant shall be capable of learning and supporting additional languages via cloud-based model updates.	4
010-010-910	1.0	3.1.7	Kiosks shall allow users to terminate sessions at any time by clicking the "Log Out" icon or via voice command to the AI Assistant.	1

ID	Ver	MRD Ref	Description	Priority*
010-010-920	1.0	3.1.2	At any time while browsing, users may purchase more time by clicking the "Buy Time" icon, via voice command, or through a proactive prompt from the AI Assistant based on usage patterns.	3
010-010-930	1.0	3.1.2	When 60 seconds of purchased time remain, the kiosk shall display a non-intrusive prompt offering additional time; the user may accept or decline. The AI Assistant may also offer personalized promotions.	1
010-010-940	1.0	3.1.7	Kiosks shall automatically terminate a user's session within 1 second of time expiration and initiate the	1

ID	Ver	MRD Ref	Description	Priority*
			secure wipe process.	
010-020	Accuracy			
010-020-010	1.0	3.1.3	All web pages and supported document types (PDF, Office docs) shall be rendered without noticeable distortion compared to their source, with the AI Assistant able to summarize or read content aloud.	1
010-020-020	1.0	3.1.3	The AI Assistant's responses to user queries shall be accurate, relevant, and comply with content policies, with continuous improvement via feedback loops.	2
010-030	Interoperability			

ID	Ver	MRD Ref	Description	Priority*
010-030-010	1.0	3.1.3	Kiosks shall allow users to access web-based email services (Gmail, Outlook, etc.) and send/receive attachments, subject to security scanning.	1
010-030-020	1.0	3.1.3	The system shall support standard web protocols (HTTP/2, HTTPS, WebSockets) and be compatible with all major cloud services.	2
010-030-030	1.0	3.1.3	Kiosks shall support printing to nearby network printers via secure release (optional).	3
010-030-040	1.0	3.1.3	The AI Support Center shall integrate with external ticketing and monitoring systems via REST APIs.	5

ID	Ver	MRD Ref	Description	Priority*
010-040	Security			
010-040-010	1.0	3.1.6	An AI-driven security suite on each kiosk shall protect against malware, ransomware, and network attacks. Detected threats shall be reported to the cloud security dashboard and automatically mitigated (e.g., session termination, kiosk lockdown).	1
010-040-020	1.0	3.1.6	An AI content-filtering agent on the kiosk shall analyze web content in real time, blocking or blurring pornographic, violent, or other objectionable material. Violations	1

ID	Ver	MRD Ref	Description	Priority*
			shall be logged and reported.	
010-040-030	1.0	3.1.6	All security events shall be logged in a centralized, immutable cloud database for analysis and auditing.	1
010-040-040	1.0	3.1.6	Security logs shall be retained for at least one year, with anonymization after 90 days for privacy.	3
010-040-050	1.0	3.1.6, 3.2.5	Upon detection of a high-severity security event (e.g., active hacking attempt), the AI agent may automatically terminate the session and lock the kiosk, alerting a human agent and local authorities if needed. The user shall receive a	2

ID	Ver	MRD Ref	Description	Priority*
			message with an appeal mechanism.	
010-040-060	1.0	N/A	AI Support Center agents shall authenticate via SSO with MFA at the start of each shift and log out at the end.	1
010-040-070	1.0	N/A	Agents may lock their session when leaving the desk; automatic timeout after 15 minutes of inactivity.	2
010-050	Compliance			
010-050-010	1.0	N/A	The system shall comply with local and international regulations regarding data protection (GDPR, CCPA), accessibility (ADA, EN 301 549), and payment security (PCI DSS).	1

020 Reliability System Requirements

The capability of the system to maintain a specified level of performance when used under specified conditions.

ID	Ver	MRD Ref	Description	Priority
020-010	Maturity			
020-010-010	1.0	3.2.1	OS and base software updates shall be tested in a staging environment and approved by the Change Control Board before being pushed to kiosks, except for critical security patches which may be expedited.	3
020-010-020	1.0	3.2.1	AI model updates (for Assistant, content filtering, etc.) shall be A/B tested on a subset of kiosks before full rollout.	3
020-010-030	1.0	3.1.6	Threat definitions and AI security models shall be updated immediately upon availability from the cloud security team.	1

ID	Ver	MRD Ref	Description	Priority
020-020	Fault-tolerance			
020-020-010	1.0	3.1.2	If network connectivity is lost during payment authorization, the kiosk shall cache the transaction and attempt to finalize upon reconnection. If the user has been charged, the session shall continue; otherwise, the transaction shall be aborted.	1
020-020-020	1.0	3.1.2	If the cloud connection is lost, the kiosk shall continue to operate in a limited offline mode (e.g., only previously cached content, no new payments) and attempt to reconnect automatically.	2
020-030	Recoverability			
020-030-010	1.0	N/A	After an unexpected crash, the kiosk shall automatically reboot and restore to the Welcome	1

ID	Ver	MRD Ref	Description	Priority
			screen within 5 minutes, with error logs sent to the cloud.	
020-040	Compliance			
020-040-010	1.0	N/A	The system shall meet or exceed industry standards for reliability (e.g., 99.5% uptime for kiosks, 99.9% for cloud services).	2

030 Usability System Requirements

The capability of the system to be understood, learned, used, and attractive to the user and the call center agents when used under specified conditions.

ID	Ver	MRD Ref	Description	Priority
030-010	Understandability			
030-010-010	1.0	3.1	The kiosk interface shall follow modern touchscreen UI conventions (large buttons, clear icons, consistent	2

ID	Ver	MRD Ref	Description	Priority
			navigation) and be intuitive for first-time users.	
030-010-020	1.0	3.1	The AI Support Center dashboard shall have a clean, role-based layout with real-time notifications and drill-down capabilities.	2
030-020	Learnability			
030-020-010	1.0	N/A	New users shall be able to start a session within 30 seconds without assistance, aided by the AI Assistant's proactive guidance.	1
030-030	Operability			
030-030-010	1.0	3.1	The kiosk user shall not need to navigate more than three screens before starting to browse.	2

ID	Ver	MRD Ref	Description	Priority
030-030-020	1.0	3.1	The AI Support Center agent shall be able to access any major function (view kiosks, view users, terminate session) within two clicks or a voice command.	2
030-040	Attractiveness			
030-040-010	1.0	3.1.1	The Welcome screen shall feature a dynamic, visually appealing design that can be updated remotely (themes, promotions).	3
030-050	Compliance			
030-050-010	1.0	N/A	The system shall comply with local handicap-access laws, including screen reader compatibility, adjustable height, and voice control.	5

040 Efficiency System Requirements

The capability of the system to provide appropriate performance, relative to the amount of resources used under stated conditions.

ID	Ver	MRD Ref	Description	Priority
040-010	Time behavior			
040-010-010	1.0	3.1.1	The kiosk shall present the Welcome screen within 60 seconds of initial power-up 95% of the time (max 120 seconds).	1
040-010-020	1.0	3.1.1	The kiosk shall reset to the Welcome screen within 10 seconds of session termination 95% of the time (max 20 seconds).	1
040-010-030	1.0	3.1.2	The first payment screen shall appear within 0.3 seconds of user interaction 95% of the time (max 1 second).	2
040-010-040	1.0	3.1.2	Payment processing (excluding external authorization) shall complete within 5 seconds 95% of the time (max 15 seconds). If external authorization exceeds 20 seconds, the user	2

ID	Ver	MRD Ref	Description	Priority
			shall be given the option to cancel.	
040-010-050	1.0	3.1.2	If payment authorization exceeds 45 seconds, the transaction shall time out, and the user may retry.	2
040-020	Resource utilization			
040-020-010	1.0	N/A	The cloud infrastructure shall support up to 10,000 concurrently active kiosks and 100 concurrent AI Support Center agents.	2
040-020-020	1.0	N/A	The system shall auto-scale cloud resources based on demand, minimizing idle capacity.	2
040-030	Compliance			
040-030-010	1.0	N/A	The system shall meet or exceed performance SLAs defined in operational contracts.	1

050 Maintainability System Requirements

The capability of the system to be modified. Modifications may include corrections, improvements, or adaptations to changes in environment, requirements, and functional specifications.

ID	Ver	MRD Ref	Description	Priority
050-010	Analyzability			
050-010-010	1.0	N/A	The development and maintenance organizations shall maintain full traceability between marketing requirements, system requirements, quality risks, test suites, and test cases using a modern ALM tool.	1
050-010-020	1.0	N/A	The system shall provide comprehensive logging and monitoring, with AI-driven anomaly detection to proactively identify potential issues.	2
050-020	Changeability			
050-020-010	1.0	N/A	The cloud architecture shall support horizontal scaling to accommodate up to 500 AI Support Center agents and 50,000 kiosks.	2

ID	Ver	MRD Ref	Description	Priority
050-020-020	1.0	N/A	Software updates shall be deployable via blue/green or canary releases to minimize downtime.	2
050-030	Stability			
050-030-010	1.0	N/A	The system shall maintain stability during updates; rollback mechanisms shall be in place for failed deployments.	1

060 Portability System Requirements

The capability of the system to be transferred from one environment to another.

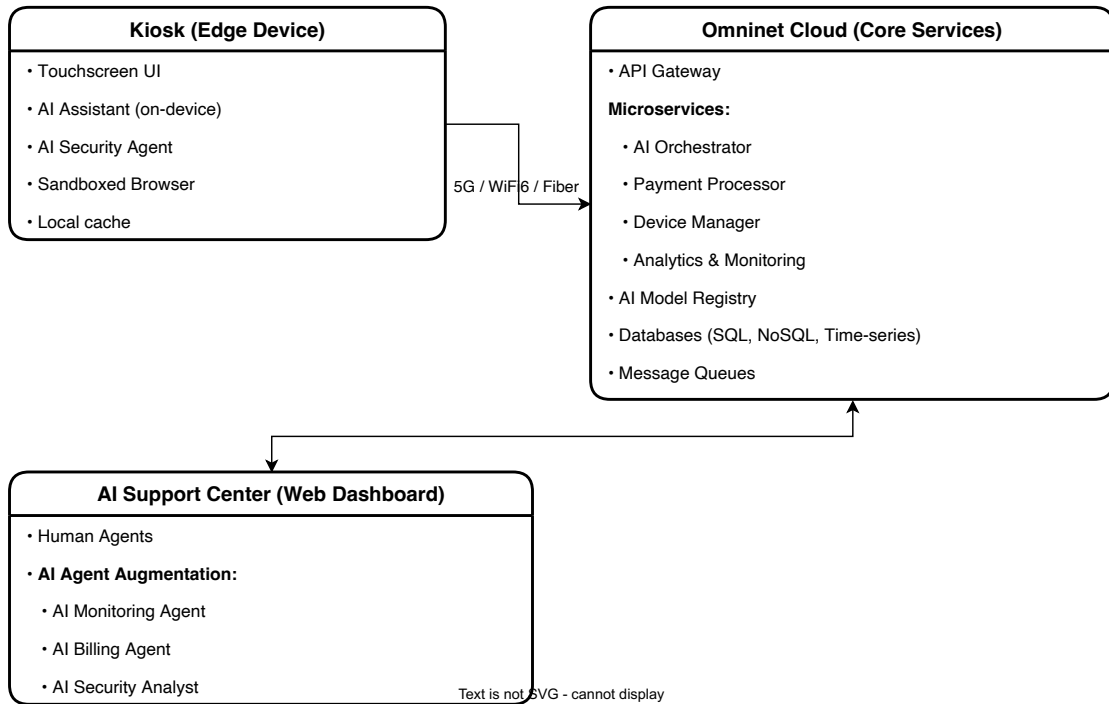
ID	Ver	MRD Ref	Description	Priority
060-010	Adaptability			
060-010-010	1.0	N/A	The kiosk hardware shall support a modern, secure OS (e.g., Windows 11 IoT Enterprise, Ubuntu Linux, or Android).	1

ID	Ver	MRD Ref	Description	Priority
060-010-020	1.0	N/A	The kiosk software shall be containerized to allow easy deployment across different hardware platforms.	2
060-020	Installability			
060-020-010	1.0	3.2.1	The cloud management system shall automatically detect out-of-date kiosk software and push updates, with the AI Orchestrator managing rollout to avoid network congestion.	1
060-030	Co-existence			
060-030-010	1.0	N/A	Kiosks shall support viewing of standard document types (PDF, DOCX, XLSX) via integrated viewers or cloud conversion services.	3
060-040	Replaceability			
060-040-010	1.0	N/A	The cloud backend shall be built using microservices	2

ID	Ver	MRD Ref	Description	Priority
			and container orchestration (Kubernetes) to allow vendor-agnostic deployment (on-prem, public cloud, hybrid).	
060-050	Compliance			
060-050-010	1.0	N/A	The kiosk web interface shall be built with standard HTML5, CSS3, and JavaScript to work on any modern browser (for remote assistance).	3
060-050-020	1.0	N/A	All software shall be developed using industry-standard languages (Python, Go, Java, C#) and frameworks with open APIs.	1

070 Design Models

070-010 Omninet AI System Architecture



070-020 Payment Processing Decision Table (Updated)

The following decision table shows the business rules governing payment processing. Payment authorization logic runs in the cloud; cash verification and NFC reading run on the kiosk.

Condition / Business Rule	1	2	3	4	5	6	7
Valid physical cash?	No	Yes	–	–	–	–	–
Valid card (chip/NFC) presented?	–	–	No	Yes	Yes	Yes	Yes
Valid PIN / biometric (if required)?	–	–	–	No	Yes	Yes	Yes
Sufficient funds / credit approved?	–	–	–	–	No	Yes	Yes
Promotional code / offer applied?	–	–	–	–	–	No	Yes

Condition / Business Rule	1	2	3	4	5	6	7
Action							
Reject cash	Yes	No	No	No	No	No	No
Reject card	No	No	Yes	Yes	No	No	No
Prompt for alternative payment	No	No	No	No	Yes	No	No
Sell requested time block	No	Yes	No	No	No	Yes	Yes
Apply promotional bonus time	No	No	No	No	No	No	Yes

070-030 Kiosk Module Flow

(A high-level data flow diagram would be inserted here, showing modules: Welcome, Payment, Browser, AI Assistant, Security, Admin Client, Cloud Sync.)

070-040 Kiosk State-Transition Diagram

(A diagram showing states: Idle/Welcome, Payment, Browsing, AddingTime, Terminated, Locked (security), Offline. Transitions triggered by user actions, timer events, AI agent decisions, or cloud commands.)

070-050 Kiosk State-Transition Table

(Table listing states, events, conditions, next state, and actions. Events: UserSelectsPayment, PaymentSuccess, TimeLow, AIFlagsThreat, AgentTerminates, NetworkLost, etc.)

070-060 Configuration Orthogonal Arrays (Sample)

(Example of a 3-factor array for testing OS, browser, and AI model version combinations.)

Config	OS (Win11/Ubuntu/Android)	Browser Engine (Chromium/Firefox)	AI Model Version (1.0/1.1/1.2)
1	Win11	Chromium	1.0
2	Win11	Firefox	1.1
3	Ubuntu	Chromium	1.1
4	Ubuntu	Firefox	1.2
5	Android	Chromium	1.2
6	Android	Firefox	1.0

Note: More comprehensive arrays would be generated for test planning.